Introduction to Econometrics

Section 2A / 2B

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What We'll Do Today

- Build good workflow habits
- Learn how to read datasets
- Set up STATA and learn some basics

Quick Introductions

You (1 min each):

- Name + pronouns
- Major
- 3 words: "Econometrics = ?"

Me:

- First-year PhD Student in Economics
- Research: Macroeconomics - savings & investment behavior
- Former data journalist

How This Section Will Work

- Be on time (we move fast)
- Use correct Zoom name (LAST NAME, FIRST NAME)
- Camera on at the start
- Expect cold-calling (low stakes!)
- Participate: Chat, Speaking, or Interrupting (I prefer to hear your voice at the bare minimum, its hard to talk to myself!)
- My office hours are 1-3 pm on Fridays.
- Feel free to mail me your questions: *pkanagar@ucsc.edu*. I respond in 24 hours.

Good Habits (Start Early)

1. **The Folder Setup**
   - /raw_data - Never overwrite your raw data!
   - /clean_data - Manipulated versions of raw data
   - /code - keep `.do` files and master scripts here
   - /output - tables, excel, plots and logs (Can make different sub-folders for each)

2. **Code should be self-explanatory**
   - Write your .do file with other people in mind.
   - Assume user knows basic STATA syntax but does not know your goals.
   - Use descriptive names:
     - Bad: `data_final_v2_revised.do`
     - **Good:** `2026-04-03_clean_wages.do`

Golden Rule: Your code is a recipe. If I have the ingredients (data), it should work in any kitchen (Reproducibility Rule).

Visualizing Best Practices

Cryptic & Unreadable

```
use "fruitdat.dta"
rename v1 fruit
label variable fruit "Type of fruit"
rename v2 sales
rename v3 price
label var sales "Sales (units)"
label var price "Price ($)"
sum price if fruit == "apple", det
```

Avoid this: No spacing, no comments, and unclear variable names.

Clean & Self-Documenting

```
// load dataset
use "fruitdat.dta"

// name variables
rename v1 fruit
rename v2 sales
rename v3 price

// label variables
label variable fruit "Type of fruit"
label var sales "Sales (units, weekly)"
label var price "Price ($)"

// average price of apples
sum price if fruit == "apple", det
```

Adopt this: *Logical sections, descriptive names, and intuitive spacing.*

Scripting & Documentation

Stata Commenting Syntax:

- `*` → Used for a **full line** comment.
- `//` → Used for notes at the **end of a line**.
- `/* ... */` → Used for large **headers or sections**.

Other Standards:

- **Self-contained:** Code runs from start to finish without manual clicks.
- **Log Files:** Always start with `log using`; it's your official record.
- Use `cd` once at the top of your code to define the project root (folder).
- **Portable Paths:** Use relative paths, not absolute paths.
  - Bad: `C:/Users/Pavitra/Documents/data.dta`
  - **Good:** `use "./data/wages.dta"`

How to read a dataset?

Checklist:

- *Unit of Observation:* Does each row represent an individual, a state, or a country-year?
- *Identify the Key:* Is there a unique, non-missing variable (like `county_fips` or `person_id`) for every row?
- *Check the Layout:* Is the data "wide" (years as columns) or "long" (years as rows)?
- *Verify the Source:* Is this the *raw* file , or a cleaned version?
- *File Format*: csv, excel or dta?

Classifying Your Data Structure:

- **Cross-section**: Many different units observed at one point in time.
- **Time series**: One single unit observed over many points in time.
- **Panel (Longitudinal)**: Many different units followed over many points in time.

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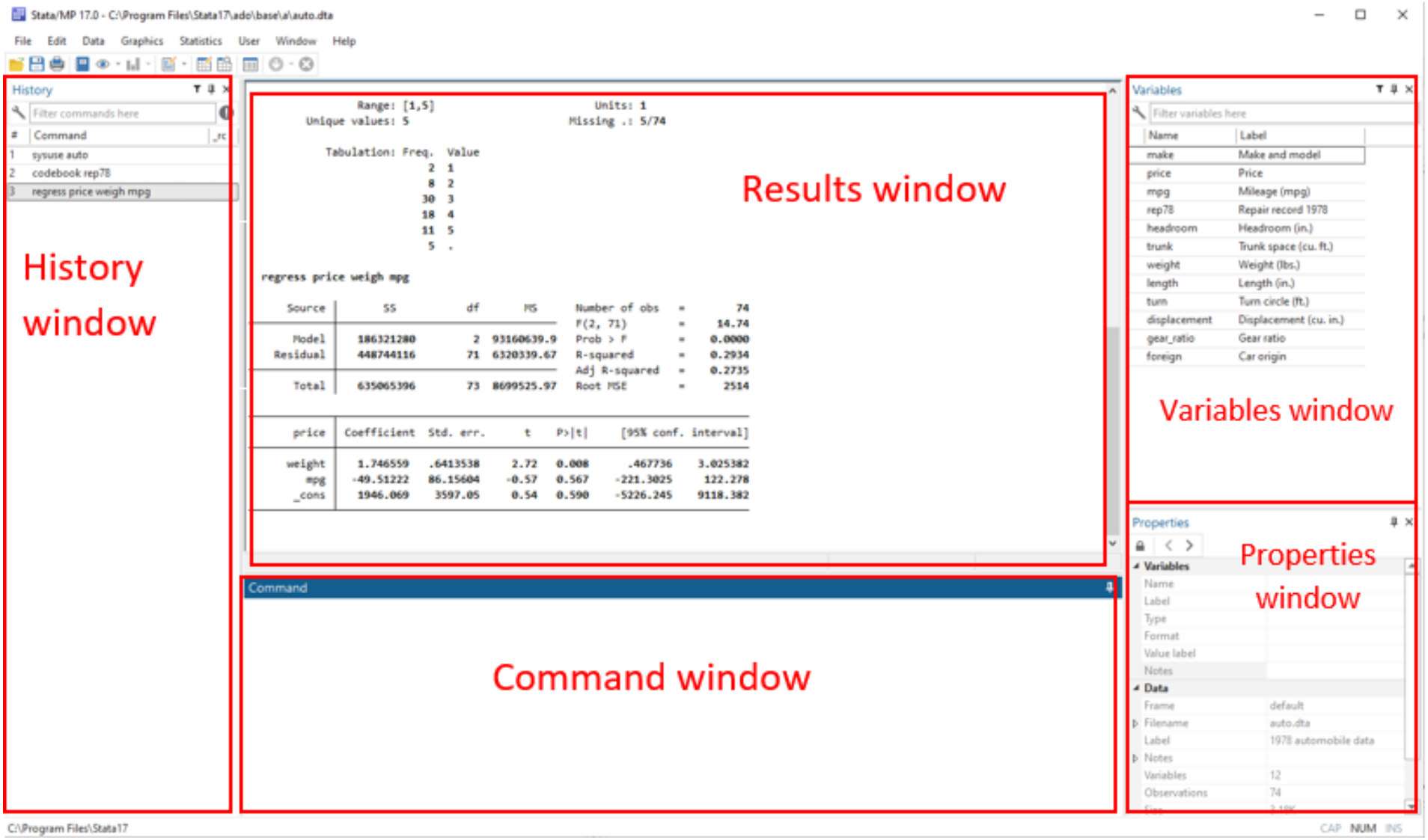
STATA Interface



STATA Interface is made up of 5 key windows

Getting Started: Do File

1. **Creating & opening scripts**
   - Type `doedit` in the Command window to create a new do file.
   - Use *File* → *Save As* to keep your "recipe" in your `/code` folder.
2. **Executing your code**
   - Highlight the lines you want to run and click the **Execute (Do)** icon.
   - **Keyboard Shortcuts:**
     - **Windows:** `Ctrl + D`
     - **Mac:** `Cmd + Shift + D`

Let's Get Live Now

- Running code in STATA, finding help
- Setting up working directory, running a log file
- Importing and examining the data
- How to create, import and save a do-file
- Data wrangling - generating new variables
- Common plots/graphs

Next Time

- Guidelines on submitting problem sets
- More STATA practice

Questions?